

Project Completion Report (PCR)

Project Name: Plutus Debugger VSCode Plugin

Project Number: 1400098

Project Start Date: Nov 24, 2025

Project Completion Date: March 31, 2026

1. Deliverables

We built and delivered a fully functional, interactive debugging and gas-profiling extension for Visual Studio Code, targeting Cardano Plutus Smart Contracts.

- **GitHub Repository (Source Code):** [Kwired/Plutus-Debugger-VSCode-Plugin](#)
- **VSCode Marketplace:** [Plutus Debugger Extension](#)
- **Open VSX Registry:** [kwired.plutus-debugger](#)
- **Documentation Portal (Single URL for Outputs):** [Plutus Debugger Docs](#)
- **Open Source Status:** Yes, released under the MIT License.
- **Testing Performed:** We conducted unit testing for the DAP protocol implementation, as well as beta testing during our live community workshop.
- **User Feedback:** Beta testers reported significant reductions in debugging time, noting that the visual ExUnit profiler was a "game changer."
- **Project Completion Video:** <https://www.youtube.com/watch?v=7rKYqu4upSo>

2. Usage

The project targets Cardano smart contract developers, DApp auditors, and educators.

- **User Count:** The extension has achieved over 500 downloads on the Open VSX Registry and has seen early adoption by several prominent ecosystem developer teams.

- **Key Actions Completed:** Developers can attach the debugger, set breakpoints, evaluate variables in real-time, and run simulated gas profiling traces.
- **Evidence of Engagement:** We hosted a live ecosystem launch workshop attended by 118 developers, with 31 survey responses yielding an average rating of 4.1/5.0. The [Community Launch Video](#) has garnered 113 views to date. Evidence documents (analytics and surveys) are included in the final submission package.

3. Impact

The measurable value created by this project is a substantial reduction in the developer friction traditionally associated with Plutus.

- **Cost or Time Savings:** Prior to this plugin, debugging Plutus required injecting trace statements, compiling, and sifting through raw node logs. The plugin reduces debugging cycles from hours to minutes.
- **Cardano Ecosystem Benefit:** By providing professional-grade tooling locally, developers deploy higher quality, gas-optimized contracts, reducing mainnet congestion and testnet failure rates.
- **Quality Proof:** We received a clean security audit report confirming zero critical vulnerabilities in the extension's architecture.

4. Sustainability

This is an ongoing project. We are committed to maintaining the VSCode plugin as the Cardano ecosystem evolves.

- **Maintenance Model:** The codebase is open-source. We will actively review community PRs and continue fixing bugs via GitHub issues.
- **Future Roadmap:** In future iterations, we plan to apply for additional funding to integrate Aiken trace support, create a standalone CLI debugger, and offer cloud-based collaborative simulation environments.
- **Permanent Storage:** All code, releases, and documentation are permanently hosted on GitHub and GitHub Pages.